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ARTÍCULO DE INVESTIGACIÓN

Resolución de la ecuación de Schrödinger mediante redes neuronales para la simulación de qubits y puertas cuánticas

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Resumen
Exploraremos la aplicación de redes neuronales informadas por la física (PINNs) para hallar la ecuación de Schrödinger, con el objetivo de simular cómo los sistemas cuánticos evolucionan con el tiempo. Se comienza con el caso general continuo y se demuestra la manera en que, en sistemas de dos niveles, la ecuación disminuye a la dinámica de un qubit. Después, se examina la manera en que las soluciones de la ecuación se asocian a transformaciones unitarias vinculadas con puertas cuánticas elementales. Por último, se sugiere una aplicación informática en la que una red neuronal aprende la función de onda $\Psi_\theta(t)$ que cumple con la ecuación de Schrödinger para un Hamiltoniano específico, lo que demuestra que las redes neuronales pueden funcionar como simuladores cuánticos aproximados.

Contenido

1. Introducción

La simulación de la dinámica cuántica es central en la física teórica y aplicada: desde la evolución de un estado ligado en un átomo hasta la operación de puertas en un computador cuántico. Tradicionalmente, dichas simulaciones se realizan mediante métodos numéricos clásicos (diferencias finitas, Crank–Nicolson, métodos espectrales como Split–Step Fourier). En los últimos años, las Physics-Informed Neural Networks (PINNs) han surgido como una alternativa para aproximar soluciones de ecuaciones diferenciales (EDOs y EDPs) imponiendo las leyes físicas directamente en la función de pérdida de una red neuronal.

2. Notacion y Consideraciones Preliminares

- $\hbar$: constante de Planck reducida
- $\psi(r, t)$: función de onda (compleja) en espacio y tiempo
- Para qubits denotaremos $|\psi(t)\rangle = (\alpha(t), \beta(t))^\top \in \mathbb{C}^2$
- Normas y productos: $(\cdot, \cdot)$ producto interno; $\|\cdot\|^2 = (\cdot, \cdot)$
- Notación de derivadas parciales: ∂_t, ∂_z, y derivadas por autograd en NNs

3. Fundamentos de la mecanica cuantica para simulacion

3.1. Espacio de Hilbert y estados cuanticos

$\mathcal{H}$ es un espacio de Hilbert si es completo con respecto a su norma. Que queremos decir que sea completo,significa que cualquier sucesion de Cauchy de elementos del espacio converge a un elemento en el mismo espacio, en tal sentido la norma de las diferencias tiende a cero.En cuanto a mecanica cuantica un conjunto fisico es descrito por un espacio complejo de Hilbert que contenga las funciones de onda para los estados posibles del conjunto

3.2. Observables,valores esperados y operadores hermiticos

3.3. Unitareidad y conservacion de probabilidad

$$\sigma_1(M) \int_\Sigma u^2 \, dA \leq \int_M |\nabla u|^2 \, dV; \tag{3.1}$$

see [1] and Section 4 below. Here ∇u is the tangential gradient of u. It is the projection of the ambient gradient $\overline{\nabla}\widetilde{u}$ on the tangent spaces of $M \subset \mathbb{R}^{n+1}$. The basic idea of our demostración is to fix a function $\widetilde{u} \in C^\infty(\mathbb{R}^{n+1})$ and consider the vector field $\overline{\nabla}\widetilde{u}$ in the ambient space $\mathbb{R}^{n+1}$. The hypersurface M is then deformed by creating "wrinkles" that tend to make the various tangent spaces T_pM, for $p \in \operatorname{int} M$, perpendicular to $\overline{\nabla}\widetilde{u}(p)$. This is achieved by "folding the surface like an accordion" in the direction perpendicular to $\overline{\nabla}\widetilde{u}$. In the limit the right-hand-side of inequality (3.1) tends to zero. Let us illustrate this strategy with a simple example.

Ejemplo 3.1. Given a smooth function $f : \overline{\mathbb{D}} \to \mathbb{R}$ vanishing on the circle $S^1 = \partial\mathbb{D}$, consider the surface

$$S_f := \text{Graph of } f = \big\{(x, y, f(x, y)) : (x, y) \in \overline{\mathbb{D}}\big\}.$$

The boundary of S_f is the same for each f. We will use the function defined by $\widetilde{u}(x, y, z) = x$ and its restriction $u = \widetilde{u}|_{S_f}$ as a trial function in inequality (3.1). Because $\nabla\widetilde{u} = (1, 0, 0)$, it follows from

Lema 4.4 that the Dirichlet energy of $u := \widetilde{u}\big|_{S_f} : S_f \to \mathbb{R}$ is given by

$$\int_{S_f} |\nabla u|^2 = \int_{\mathbb{D}} \frac{1 + f_y^2}{\sqrt{1 + f_x^2 + f_y^2}}\, dxdy.$$

For $n \in \mathbb{N}$, define $f = f_n : \overline{\mathbb{D}} \to \mathbb{R}$ by

$$f(x, y) = \sin(nx)\overbrace{(1 - x^2 - y^2)}^{\phi(x,y)}.$$

It follows from

$$f_x^2 = n^2\Big(\cos(nx)\phi + \frac{1}{n}\sin(nx)\phi_x\Big)^2 \quad \text{and} \quad f_y^2 = \sin^2(nx)\phi_y^2$$

that

$$\lim_{n\to\infty} \int_{S_{fn}} |\nabla u|^2 = 0.$$

Together with (3.1), this shows that $\lim_{n\to\infty} \sigma_1(S_{f_n}) = 0$.

La demostración del Teorema **??** is based on the above idea, but it is technically more involved, because we want to localize this argument to a small neighbourhood of a point p of the boundary. This is a significant gain compared to the above example, because it allows the construction of an arbitrary finite number of disjointly supported trial functions with small Dirichlet energy, leading to the collapse of each eigenvalue σ_k rather than just σ_1. For the sake of readability and simplicity, the deformations that we use in the demostración del Teorema **??** are Lipschitz continuous but only piecewise smooth.

Plan del Artículo

In Section 4 we review the min-max characterization of Steklov eigenvalues, and we prove a lemma regarding the control of the Dirichlet energy under quasi-isometries. We then proceed to construct the perturbed hypersurfaces in Section 5. We use a quasi-isometric chart to a hypersurface with a flat boundary. The perturbed submanifold is then constructed by considering the graph of a locally supported oscillating function. Finally, in Section 6 an appropriate trial function is used to conclude the demostración del Teorema **??**.

4. Notación y Consideraciones Preliminares

Let M be a smooth compact manifold with boundary Σ. The volume form on M is written dV, while the volume form on Σ is dA. We denote by $H^1(M)$ the standard Sobolev space of functions in $L^2(M, dV)$ with weak first derivative in $L^2(M, dV)$. The Steklov eigenvalues σ_k admits a variational characterization in terms of the *Steklov–Rayleigh quotient* of a function $0 \neq u \in H^1(M)$,

$$\mathcal{R}(u) = \frac{\int_M |\nabla u|^2\, dV}{\int_\Sigma u^2\, dA}.$$

The numerator $D(u) = \int_M |\nabla u|^2\, dV$ is the *Dirichet energy* of $u \in H^1(M)$. It is well known that

$$\sigma_k = \min_{\substack{S\subset H^1(M)\\ \dim S = k+1}} \max_{u\in S\setminus\{0\}} \mathcal{R}(u), \tag{4.1}$$

where the minimum is taken over all $(k+1)$-dimensional linear subspaces $S \subset H^1(M)$.

4.1. Quasi-isometries and Dirichlet Energy

Let M and $\widetilde{M}$ be two n-dimensional Riemannian manifolds with boundary. A diffeomorphism $\phi\colon M \to \widetilde{M}$ is a *quasi-isometry* with constant $C \geq 1$ if for each $p \in M$ and each $0 \neq v \in T_pM$,

$$\frac{1}{C} \leq \frac{\|D_p\phi(v)\|^2}{\|v\|^2} \leq C.$$

Quasi-isometries provide a control of the Dirichlet energy of a function.

Lema 4.1. *Let $\phi\colon M \to \widetilde{M}$ be a quasi-isometry with constant $C \geq 1$. Let $f \in H^1(\widetilde{M})$; then*

$$\frac{1}{C^{\frac{n}{2}+1}} \leq \frac{\|\nabla(f\circ\phi)\|^2_{L^2(M)}}{\|\nabla f\|^2_{L^2(\widetilde{M})}} \leq C^{\frac{n}{2}+1}.$$

Demostración. Let $\widetilde{g}$ be the Riemannian metric of $\widetilde{M}$ and let g be that of M. Let $\widehat{g} = \phi^\star(\widetilde{g})$ be the pull-back of the metric $\widetilde{g}$. Because ϕ is a quasi-isometry with constant C, the following holds for each $0 \neq v \in TM$,

$$\frac{1}{C} \leq \frac{g(v,v)}{\widehat{g}(v,v)} \leq C.$$

It follows that

$$\frac{1}{C}g(\nabla_g f, \nabla_g f) \leq \widehat{g}(\nabla_{\widehat{g}} f, \nabla_{\widehat{g}} f) \leq Cg(\nabla_g f, \nabla_g f).$$

The corresponding volume forms satisfy

$$C^{-n/2}dV_g \leq dV_{\widehat{g}} \leq C^{n/2}dV_g.$$

This leads to

$$\begin{aligned}
\|\nabla(f\circ\phi)\|^2_{L^2} &= \int_M g\big(\nabla_g(f\circ\phi), \nabla_g(f\circ\phi)\big)\, dV_g \\
&\leq C^{n/2+1}\int_M \widehat{g}\big(\nabla_{\widehat{g}}(f\circ\phi), \nabla_{\widehat{g}}(f\circ\phi)\big)\, dV_{\widehat{g}} \\
&= C^{n/2+1}\int_{\widetilde{M}} \widetilde{g}(\nabla_{\widetilde{g}} f, \nabla_{\widetilde{g}} f)\, dV_{\widetilde{g}}.
\end{aligned}$$

The proof of the lower bound is identical, and accordingly omitted. □

4.2. Quasi-isometric Charts

Recall that a subset $M \subset \mathbb{R}^{n+1}$ is a hypersurface with boundary if for each $p \in M$, there exist open sets $W, W' \subset \mathbb{R}^{n+1}$ with $p \in W$ and a diffeomorphism $\psi : W \to W'$ such that $\psi(M\cap W)$ is an open set in the half-space

$$H = \{x \in \mathbb{R}^{n+1} : x_{n+1} = 0, x_1 \geq 0\}.$$

The point $p \in M$ is on the boundary Σ of M if and only if ψ sends it to the boundary of the half-space H:

$$\psi(p) \in \partial H := \{x \in H : x_1 = 0\}.$$

This definition is coherent. It does not depend on the choice of the diffeomorphism ψ; see [2, Chapter 1] for details. By further restricting ψ and scaling if necessary, we can assume that it is a quasi-isometry and that its image W' is a cylinder. This is summed up in the next lemma.

Lema 4.2. *For each $p \in \Sigma$, there exists a quasi-isometry*

$$\psi : W \longrightarrow W' = B_{\mathbb{R}^n}(0,1) \times (-1,1)$$

with $\psi(p) = 0$ and such that the image of $M \cap W$ is

$$U := \psi(M \cap W) = \{x \in H : |x| < 1\}.$$

Observación 4.3. We identify $U \subset \mathbb{R}^{n+1}$ with a subset of $\mathbb{R}^n$ so that we can write $x = (x_1, \dots, x_n) \in U$ instead of $x = (x_1, \dots, x_n, 0) \in U$.

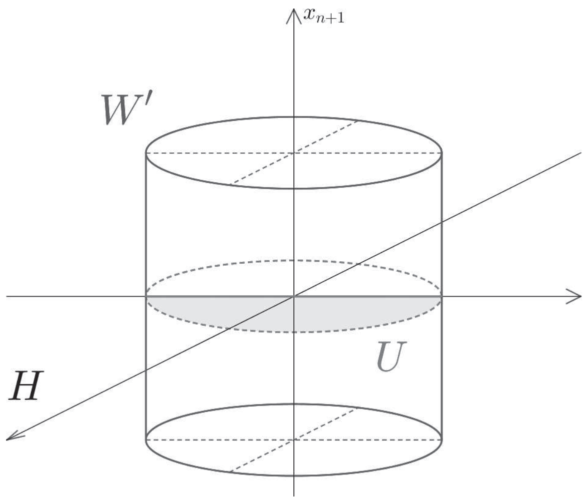

Figura 1. *The domain U.*

In particular, the restriction of ψ to $\Sigma \cap W$ is also a quasi-isometry.

4.3. Dirichlet Energy on the Graph of a Function

Let $U \subset \mathbb{R}^n$ be a bounded open set and let $f : U \to \mathbb{R}$ be a bounded smooth function. Consider the graph

$$S_f = \{(x, f(x)) : x \in U\} \subset \mathbb{R}^{n+1}.$$

Given a function $u : U \to \mathbb{R}$, define $\widetilde{u} : U \times \mathbb{R} \to \mathbb{R}$ by $\widetilde{u}(x, x_{n+1}) = u(x)$ and define $u_f : S_f \to \mathbb{R}$ by

$$u_f(x, f(x)) = u(x) = \widetilde{u}\big|_{S_f}. \tag{4.2}$$

Lema 4.4. *The Dirichlet energy of $u_f : S_f \to \mathbb{R}$ is*

$$\int_{S_f} |\nabla u_f|^2 \, dV = \int_U \frac{|\nabla u|^2 + |\nabla u|^2 |\nabla f|^2 - \langle \nabla u, \nabla f \rangle^2}{\sqrt{1 + |\nabla f|^2}} \, dx, \tag{4.3}$$

where on the left-hand-side ∇, dV, and the norm are taken on S_f, and on the right-hand-side $dx = dx^1 \cdots dx^n$ is the Lebesgue measure on U, while ∇ is the usual gradient on $\mathbb{R}^n$.

The Cauchy–Schwarz inequality gives $|\nabla u|^2|\nabla f|^2 - \langle \nabla u, \nabla f\rangle^2 \geq 0$ with equality if and only if $\nabla u = c\nabla f$ for some constant c.

Demostración del Lema 4.4. To simplify notation, we will write $S = S_f$. For any point $p \in S$, the gradient $\nabla u_f \in T_pS$ is the projection of $\overline{\nabla}\widetilde{u}$ on T_pS, that is,

$$\nabla u_f = \overline{\nabla}\widetilde{u} - \langle \overline{\nabla}\widetilde{u}, N\rangle N,$$

where N is a unit normal vector to T_pS. It follows from

$$\nabla\widetilde{u} = \Big(\frac{\partial u}{\partial x_1}, \ldots, \frac{\partial u}{\partial x_n}, 0\Big),$$
$$N = \frac{1}{\sqrt{1 + |\nabla f|^2}}\Big(e_{n+1} - \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} e_i\Big).$$

that

$$|\nabla u_f|^2 = \frac{|\nabla u|^2 + |\nabla u|^2|\nabla f|^2 - \langle \nabla u, \nabla f\rangle^2}{1 + |\nabla f|^2}. \tag{4.4}$$

The volume element on S_f is given by

$$dV = \sqrt{1 + |\nabla f|^2}. \tag{4.5}$$

Together with identity (4.4), this completes the proof. □

5. Perturbation of the Submanifold M

Given $p \in \Sigma$, let ψ be the quasi-isometric chart provided by Lema 4.2. In order to prove Teorema **??**, we will deform the submanifold M in the neighborhood W of the point p by deforming the neighborhood $U \subset W'$ inside W' and pulling back to W using the quasi-isometry ψ. Consider a smooth function $f : U \longrightarrow \mathbb{R}$ that is supported in the interior of U and which satisfies $|f(x)| < 1$ for each $x \in U$. This last condition implies that the graph of f,

$$S_f = \{(x, f(x)) : x \in U\},$$

is contained in the cylinder W'. Hence it can be used to define a deformation of M as follows:

$$\widetilde{M}_f := (M \setminus W) \cup \psi^{-1}(S_f). \tag{5.1}$$

Observación 5.1. It might be possible to prove Teorema **??** by performing a deformation of M directly in the ambient space $\mathbb{R}^{n+1}$, but it appears to be simpler to use quasi-isometric charts.

5.1. Deformation Function

We now construct specific functions f and u such that the Dirichlet energy of u_f, defined by (4.2), is small. Our method is based on Lema 4.4, which shows that if ∇u and ∇f are parallel the numerator of (4.3) is independent of ∇f, while the denominator behaves as $|\nabla f|$. Thus, we want f and u to have parallel gradients with $|\nabla f|$ big to get a small Dirichlet energy for u_f.

Consider numbers $\epsilon, \partial_1, \partial_2, \rho > 0$ that are sufficiently small and define $\partial := \partial_1 + \partial_2$. These constants will be adjusted later in equation (6.6). Let $q = (\partial, 0, \ldots, 0) \in H$. Consider the following subsets of U:

$$\begin{aligned}
A &:= \big\{(x_1, \ldots, x_n) \mid x_1 \geq \partial, \|x - q\| \leq \epsilon\big\}, \\
B &:= \big\{(x_1, \ldots, x_n) \mid \partial_1 \leq x_1 \leq \partial, \|\Pi x\| \leq \epsilon\big\}, \\
C &:= \big\{(x_1, \ldots, x_n) \mid 0 \leq x_1 \leq \partial_1, \|\Pi x\| \leq \epsilon\big\}, \\
D &:= \big\{x \in U \setminus (A \cup B \cup C) \mid d(x, A \cup B \cup C), \leq \rho\big\}
\end{aligned}$$

where $\Pi(x_1, x_2, \ldots, x_n) = (x_2, \ldots, x_n)$ is the projection on the boundary. Let $\Omega = A \cup B \cup C$. Vea la Figura 2, where the x_1-axis is vertical.

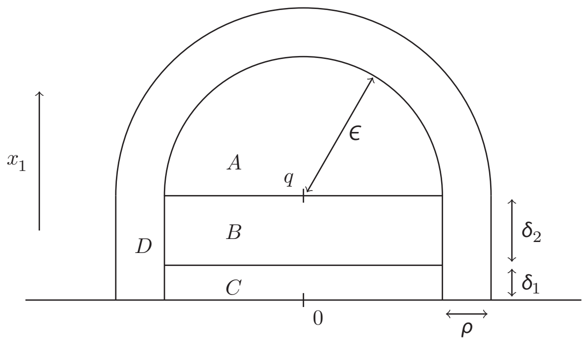

Figura 2. *Perturbation region.*

Define cutoff functions

$$\eta\colon [0, \infty) \longrightarrow \mathbb{R}, \qquad \gamma\colon [0, \partial] \longrightarrow \mathbb{R},$$

$$\eta(t) = \max\{0, 1 - t\}, \qquad \gamma(t) = \begin{cases} 0 & \text{if } t \leq \partial_1, \\ t - \partial_1 & \text{if } \partial_1 \leq t \leq \partial. \end{cases}$$

Given $\omega > 0$, define $f\colon U \longrightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} \eta\big(\frac{d(x,\Omega)}{\rho}\big)\gamma(x_1)F(\omega\|\Pi x\|) & \text{if } x_1 \leq \partial, \\ \eta\big(\frac{d(x,\Omega)}{\rho}\big)\partial_2 F(\omega\|x - q\|) & \text{if } x_1 \geq \partial. \end{cases} \tag{5.2}$$

6. Trial Function

The trial function u is supported on $\Omega \subset U$ and is defined by

$$u(x) = \begin{cases} 1 - \frac{\|\Pi x\|}{\epsilon} & \text{if } x \in B \cup C, \\ 1 - \frac{\|x - q\|}{\epsilon} & \text{if } x \in A, \\ 0 & \text{elsewhere.} \end{cases} \tag{6.1}$$

By construction, the function u_f defined by (4.2) belongs to $H^1(S_f)$, and we can estimate its Dirichlet energy. On A, the Dirichlet energy of u_f can be made small by taking ω big. Indeed, for almost all $x \in A$,

$$\nabla f = \pm \partial_2 \omega \frac{x-q}{\|x-q\|}, \tag{6.2}$$

$$\nabla u = -\frac{1}{\epsilon} \frac{x-q}{\|x-q\|}, \tag{6.3}$$

and using the fact that ∇f and ∇u are parallel, the Dirichlet energy is

$$\begin{aligned}\int_{S_f \cap A \times \mathbb{R}} |\nabla u_f|^2 dV &= \int_A \frac{|\nabla u|^2}{\sqrt{1+|\nabla f|^2}}\, dx = \frac{1}{\epsilon^2} \frac{1}{\sqrt{1+\partial_2^2 \omega^2}} \operatorname{Vol} A \\ &= \frac{c_1 \epsilon^{n-2}}{\sqrt{1+\partial_2^2 \omega^2}}\end{aligned}$$

where c_1 is some dimensional constant.

On B and C, ∇f and ∇u are not parallel, but it is possible to make the Dirichlet energy small by making the volume of B and C small. For B, we have for almost all $x \in B$:

$$\nabla f(x) = F(\omega \|\Pi x\|) e_1 \pm \gamma(x_1) \omega \frac{\Pi x}{\|\Pi x\|}, \tag{6.4}$$

$$\nabla u(x) = -\frac{1}{\epsilon} \frac{\Pi x}{\|\Pi x\|}, \tag{6.5}$$

and since e_1 and Πx are orthogonal,

$$|\nabla f(x)|^2 = F(\omega \|\Pi x\|)^2 + \gamma(x_1)^2 \omega^2.$$

Then the Dirichlet energy on B is

$$\begin{aligned}\int_{S_f \cap B \times \mathbb{R}} |\nabla u_f|^2 dV &= \int_B \frac{\frac{1}{\epsilon^2} + \frac{1}{\epsilon^2} F(\omega\|\Pi x\|)^2}{\sqrt{1 + F(\omega\|\Pi x\|)^2 + \gamma(x_1)^2 \omega^2}} dx \\ &\le \frac{1}{\epsilon^2} \int_B \frac{2}{\sqrt{1+\gamma(x_1)^2\omega^2}} dx \\ &= c_2 \epsilon^{n-3} \int_0^{\partial_2} \frac{1}{\sqrt{1+x_1^2\omega^2}} dx_1 \\ &= c_2 \epsilon^{n-3} \frac{\ln(\partial_2 \omega + \sqrt{1+\partial_2^2})}{\omega},\end{aligned}$$

where c_2 is a constant that depends only on the dimension. And on C, since $f = 0$, the Dirichlet energy is simply

$$\int_{S_f \cap C \times \mathbb{R}} |\nabla u_f|^2 dV = \frac{1}{\epsilon^2} \operatorname{Vol} C = c_3 \partial_1 \epsilon^{n-3},$$

where c_3 is a constant that depends only on the dimension. The denominator in the Steklov–Rayleigh quotient of u_f satisfies

$$\int_{B(0,\epsilon)} \left(1 - \frac{|x|}{\epsilon}\right)^2 dx \geq \frac{1}{4} \operatorname{Vol}\left(B\left(0, \frac{\epsilon}{2}\right)\right) = c_4 \epsilon^{n-1},$$

for some constant $c_4 > 0$. In total, the Steklov–Rayleigh quotient of u_f on S_f is bounded as follows:

$$\begin{aligned}\mathcal{R}(u_f) &\leq \frac{1}{c_4 \epsilon^{n-1}} \left(c_1 \frac{\epsilon^{n-2}}{\sqrt{1 + \partial_2^2 \omega^2}} + c_2 \frac{\epsilon^{n-3} \ln(\partial_2 \omega + \sqrt{1 + \partial_2^2 \omega^2})}{\omega} + c_3 \partial_1 \epsilon^{n-3} \right) \\ &= \frac{1}{c_4} \left(c_1 \frac{\epsilon^{-1}}{\sqrt{1 + \partial_2^2 \omega^2}} + c_2 \frac{\epsilon^{-2} \ln(\partial_2 \omega + \sqrt{1 + \partial_2^2 \omega^2})}{\omega} + c_3 \partial_1 \epsilon^{-2} \right).\end{aligned}$$

We are now ready to define the constants more precisely. By using the following:

$$\partial_1 = \epsilon^3, \quad \partial_2 = \epsilon^{3/2}, \quad \omega = \epsilon^{-3}, \quad \rho = \epsilon, \tag{6.6}$$

we obtain

$$\mathcal{R}(u_f) = \mathcal{O}(\epsilon^{1/2}) \quad \text{as } \epsilon \longrightarrow 0.$$

We have proved that a local perturbation of U allows the construction of a local trial function with arbitrarily small Steklov–Rayleigh quotient. The proof of our main result is now an easy consequence.

Demostración del Teorema **??**. Without loss of generality, we work under the asumption that $k \in \mathbb{N}$ is fixed. The general case will then follow by a standard diagonal selection argument. Let $k, j \in \mathbb{N}$ with j sufficiently large. Let $p_1, \ldots, p_{k+1} \in B(p, \frac{1}{j}) \cap \Sigma$ be distinct points, and let ψ be the quasi-isometric chart from Lema 4.2. For each p_i, we follow the above construction to obtain deformation functions f_i that are disjointly supported, by taking $\epsilon > 0$ small enough, and trial functions u_i that have disjoint supports contained in in $\psi(B(p, \frac{1}{j}) \cap M)$. By possibly choosing a smaller ϵ in the previous construction, we guarantee that the Rayleigh quotient of each u_i is smaller than $1/j$. Consider the deformation function $f = f_1 + \cdots + f_{k+1}$ supported in $B(p, 1/j)$ and the perturbed manifold $M_j = \widetilde{M}_f$. Taking the pullback by ψ, we obtain $k+1$ trial functions $\psi^*(u_i)$ with disjoint supports and from Lema 4.1, their Rayleigh quotient is less than c/j where c is a constant depending on ψ. By the variational characterization (4.1) of the eigenvalue σ_k, we conclude that $\sigma_k(M_j) \leq c/j$.

Note that

$$\operatorname{Vol}(S_f) = \int_{\Omega \cup D} \sqrt{1 + |\nabla f|^2} dx_1 \cdots dx_n \geq \operatorname{Vol}(\Omega \cup D).$$

Tabla 1. *Este es el título de la tabla..*

	Element 1			Element 2*		
Projectile	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40

*Este es un ejemplo de nota al pie de tabla.

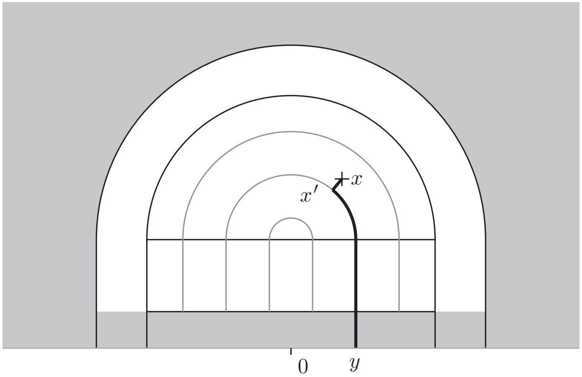

Figura 3. *The set $\{x \mid f(x) = 0\}$ is shown in grey. The path in bold from an arbitrary $x \in \Omega$ to a point on $\partial \widetilde{M}_f$ has length going to 0 as $\epsilon \to 0$.*

It follows from (6.2) and (6.4) that, on Ω, the following holds:

$$|\nabla f|^2 \leq 1 + \partial_2^2 \omega^2 = \epsilon^{-3}.$$

Similarly, it follows from (5.2) that on D,

$$|\nabla f|^2 \leq c_5 \left(\frac{\partial_2^2}{\epsilon^2} + \partial_2^2 \omega^2 \right) = c_5(\epsilon + \epsilon^{-3}),$$

where c_5 is a positive constant. It follows that

$$\begin{aligned} |\operatorname{Vol}(S_f) - \operatorname{Vol}(\Omega \cup D)| &= \int_{\Omega \cup D} \sqrt{1 + |\nabla f|^2} dx_1 \cdots dx_n - \operatorname{Vol}(\Omega \cup D) \\ &= \mathcal{O}(\epsilon^{n-3/2}), \end{aligned}$$

which goes to 0 for $n \geq 2$. Finally, it is clear that the curvatures of $\partial \widetilde{M}_f$ do not change, as M is kept fixed on some neighborhood of the boundary due to the localisation of f by the function γ, which vanishes near the boundary; see Figure 3 and the definition (5.2) of f. □

7. Conclusiones

Algunas conclusiones aquí.

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Conflictos de intereses. A statement about any financial, professional, contractual or personal relationships or situations that could be perceived to impact the presentation of the work — or 'None' if none exist

Declaración de disponibilidad de datos. A statement about how to access data, code and other materials allowing users to understand, verify and replicate findings — e.g. Replication data and code can be found in Harvard Dataverse: https://doi.org/link.

Estándares éticos. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Contribuciones de los autores. A.A. and A.B.C. designed the study, abstracted the data wrote the first draft, and approved the final version of the manuscript. A.R.E.J., M.R.L., K.L.S., and A.D.P. revised the manuscript and approved the final version.

Material suplementario. State whether any supplementary material intended for publication has been provided with the submission.

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